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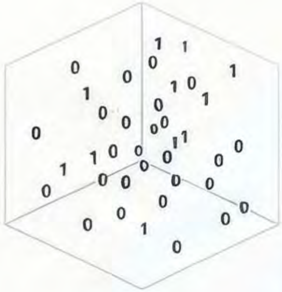
# Word Embedding: The Geometry of Meaning

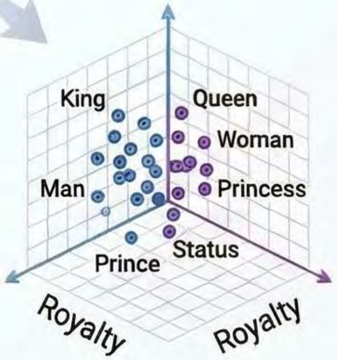
Based on the research of Dr Partha Majumdar, Department of Computer Science, Swiss School of Business and Management (SSBM), Geneva, Switzerland. (ORCID: 0009-0002-7375-8034).

## From Sparse Vectors to Semantic Space



### The Failure of One-Hot Encoding

Traditional sparse vectors are “semantically blind,” resulting in 0.0 similarity between related words.



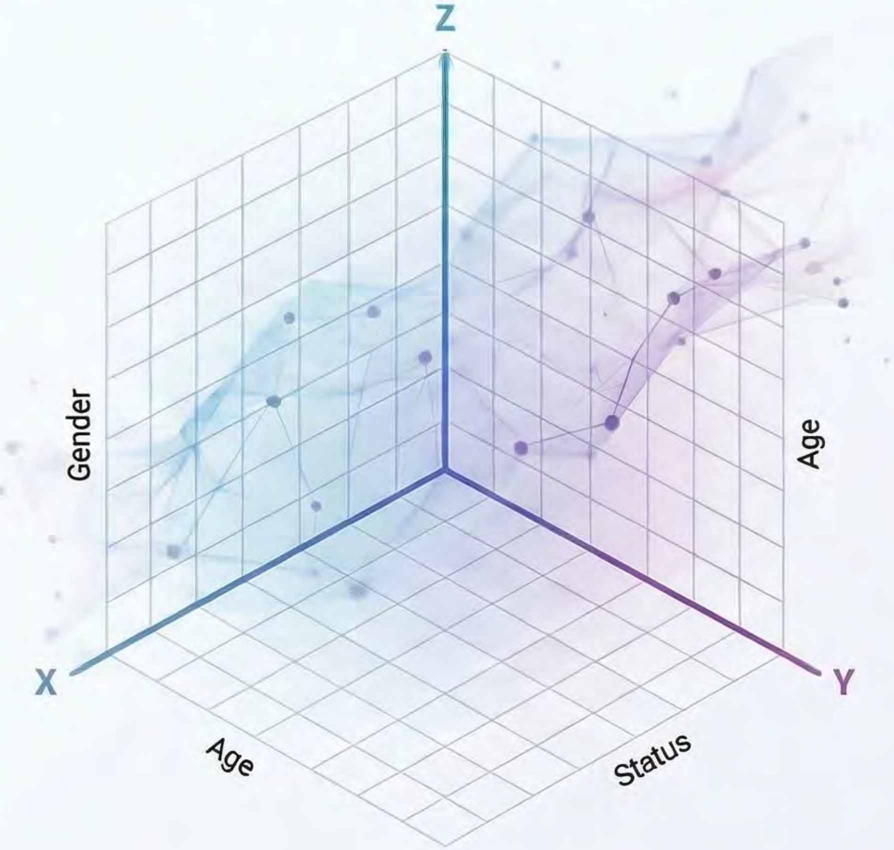

### High-Dimensional Word Vectors

Modern embeddings map words to primary semantic axes like Age, Gender, and Royalty.

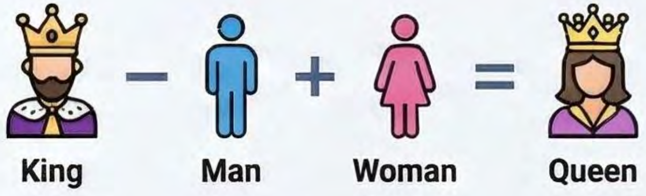
### The Curse of Dimensionality

Replacing massive, inefficient vocabularies with dense, structured 3D semantic coordinates.

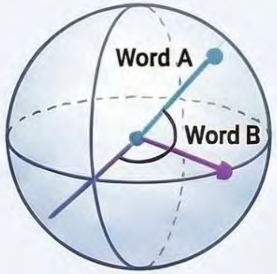
## Manifold of Meaning




## The Logic of Vector Arithmetic




Mathematical transformations allow models to solve linguistic analogies through spatial movement.

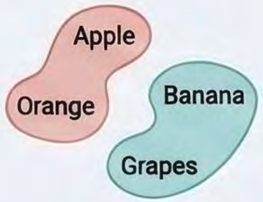



### Cosine Similarity

A geometric compass measuring semantic alignment via vector angles rather than magnitude.

### Semantic Clustering

Using t-SNE or PCA to project related concepts into visual "semantic islands."



# The One-Hot Encoding Dilemma: Why Simple Vectors Fail

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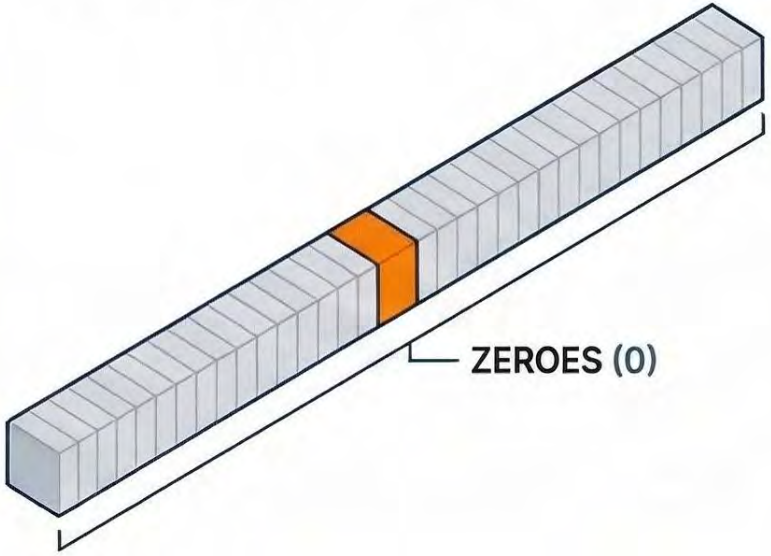
## THE SPARSE VECTOR STRUCTURE

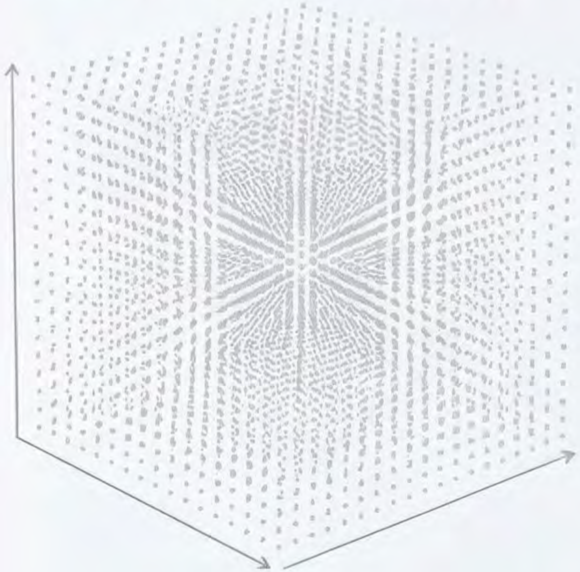



**High-Dimensional Word Vectors:**
Vector size matches vocabulary size. Example: 100,000 dimensions for 100,000 words.

**The "Hot" Bit Representation:**
Every word vector contains exactly one "1" while all other elements are zero.

## MATHEMATICAL & COMPUTATIONAL FAILURES

### The Curse of Dimensionality

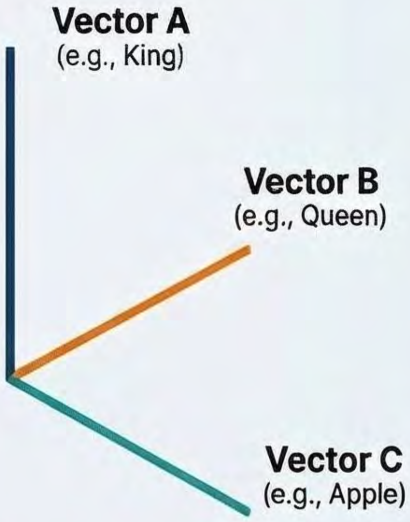


Computational Inefficiency

Extreme Memory Requirements

High Risks of Model Overfitting

### Semantic Blindness via Orthogonality




All vectors are perfectly perpendicular, making related words mathematically indistinguishable from unrelated ones.

**Word Pair Comparison**

**King vs. Queen** 0.0
Cosine Similarity: 0.0 | Relationship Detected: None

**Apple vs. Pear** 0.0
Cosine Similarity: 0.0 | Relationship Detected: None

**Cat vs. Dog** 0.0
Cosine Similarity: 0.0 | Relationship Detected: None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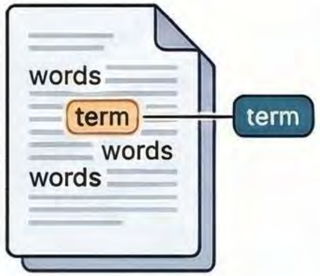
# Understanding TF-IDF: Frequency-Based Weighting for NLP

TF-IDF (Term Frequency-Inverse Document Frequency) balances local word frequency against corpus-wide rarity to identify mathematically distinct words. While effective for basic classification, it shares structural inefficiencies with one-hot encoding.

## THE MATHEMATICAL CALCULATION

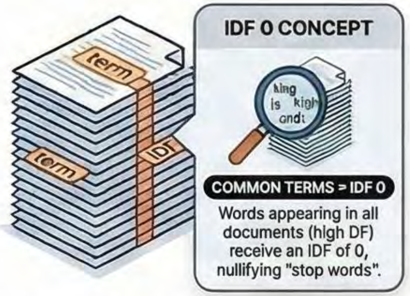
### 1. Term Frequency (TF)




Normalises word frequency by dividing word count in a document by the total words.

$$TF(w, d) = \frac{Count(w, d)}{Total\ words\ in\ d}$$

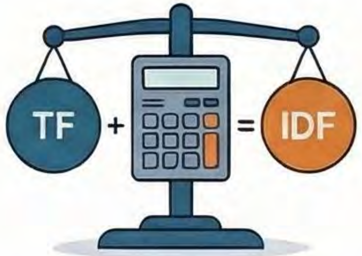
### 2. Inverse Document Frequency (IDF)




Calculated as log(N/DF), where N is total documents and DF is documents containing the word.

$$IDF(w) = \log\left(\frac{N}{DF(w)}\right)$$

### 3. TF-IDF Weighting




The final vector is the product of TF and IDF (TF-IDF = TF * IDF).

$$TF\text{-}IDF = TF(w, d) * IDF(w)$$

## CORE FORMULA REFERENCE

| | | |
|---|---|---|
| **Term Frequency** | $TF(w, d) = \frac{Count(w, d)}{Total\ words\ in\ d}$ | w = word; d = document |
| **Inverse Document Frequency** | $IDF(w) = \log\left(\frac{N}{DF(w)}\right)$ | N = total docs; DF = docs with word |
| **Final Weight** | $TF\text{-}IDF = TF(w, d) * IDF(w)$ | - |

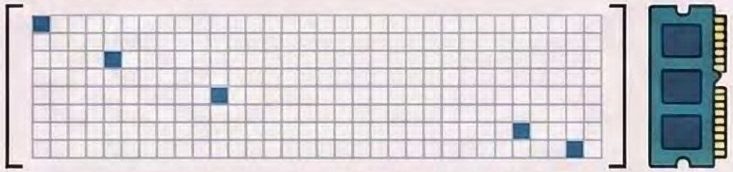
## KEY FEATURES AND LIMITATIONS


### Natural Suppression of Common Terms

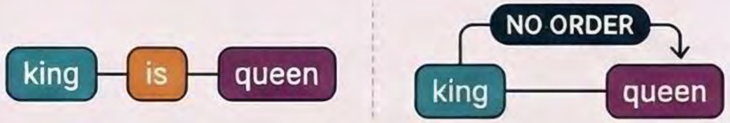
Words appearing in all documents (high DF) receive an IDF of 0, nullifying "stop words".

### ⚠ LIMITATIONS

#### Sparse, High-Dimensional Vectors



Vector sizes match vocabulary size, leading to extreme memory requirements and computational inefficiency.




#### Semantic Blindness & Structural Issues

Fails to encode relationships between terms and completely ignores the sequential order of words.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# TF-IDF Representation and Document Similarity Analysis

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## Phrases and Their Unique Words

| Document | Phrase | Unique Words |
|---|---|---|
| D1 | king rules the royal kingdom | king, kingdom, royal, rules, the |
| D2 | queen rules the royal palace | palace, queen, royal, rules, the |

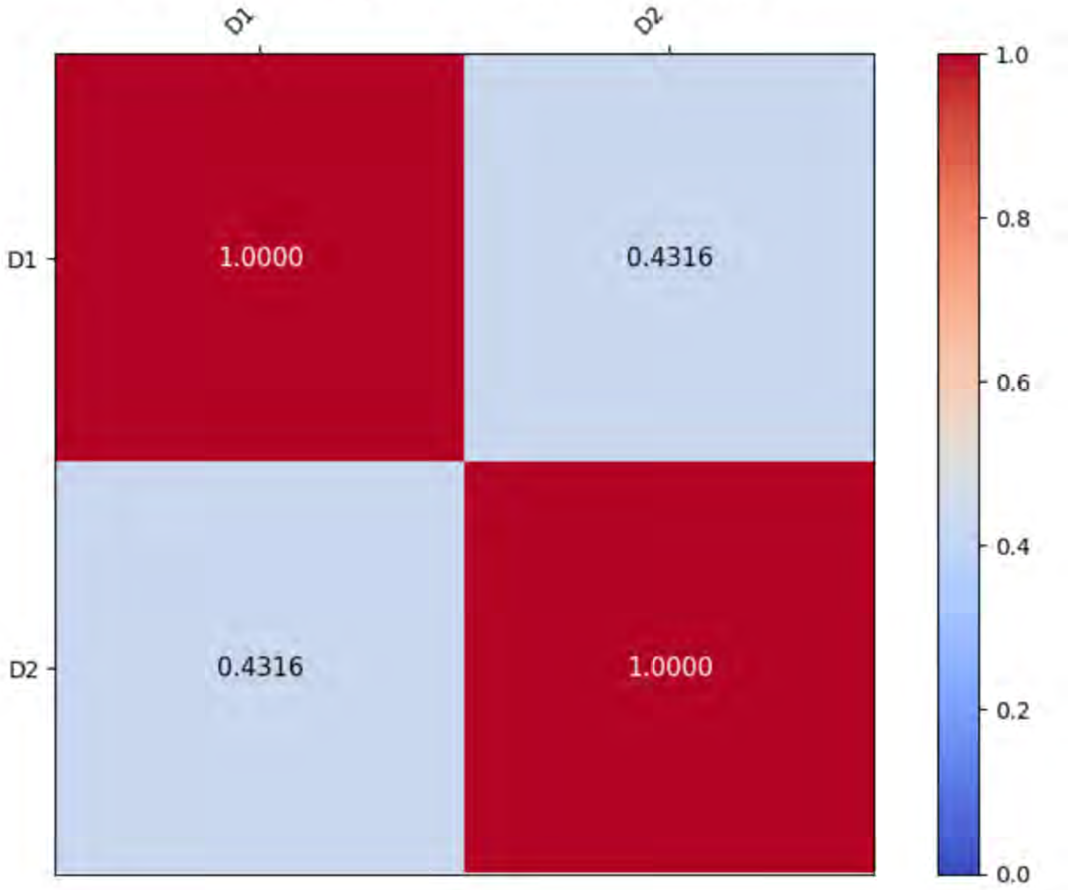
## TF-IDF Calculations for Each Word in Each Document

| Document | Word | Count | TF | DF | IDF | TF-IDF |
|---|---|---|---|---|---|---|
| D1 | king | 1 | 0.2000 | 1 | 0.6931 | 0.1386 |
| D1 | kingdom | 1 | 0.2000 | 1 | 0.6931 | 0.1386 |
| D1 | palace | 0 | 0.0000 | 1 | 0.6931 | 0.0000 |
| D1 | queen | 0 | 0.0000 | 1 | 0.6931 | 0.0000 |
| D1 | royal | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |
| D1 | rules | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |
| D1 | the | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |
| D2 | king | 0 | 0.0000 | 1 | 0.6931 | 0.0000 |
| D2 | kingdom | 0 | 0.0000 | 1 | 0.6931 | 0.0000 |
| D2 | palace | 1 | 0.2000 | 1 | 0.6931 | 0.1386 |
| D2 | queen | 1 | 0.2000 | 1 | 0.6931 | 0.1386 |
| D2 | royal | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |
| D2 | rules | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |
| D2 | the | 1 | 0.2000 | 2 | 0.0000 | 0.0000 |

## TF-IDF Document-Term Matrix

| Document | Phrase | king | kingdom | palace | queen | royal | rules | the |
|---|---|---|---|---|---|---|---|---|
| D1 | king rules the royal kingdom | 0.1386 | 0.1386 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| D2 | queen rules the royal palace | 0.0000 | 0.0000 | 0.1386 | 0.1386 | 0.0000 | 0.0000 | 0.0000 |

## Cosine Similarity Between Documents (TF-IDF)





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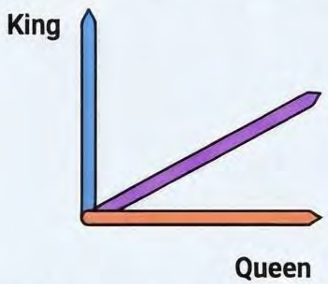
# The Geometry of Meaning:
## Understanding Vector Arithmetic and Analogical Reasoning

### The Failure of Simple Vectors




**The Orthogonality Problem**

Traditional vectors (One-Hot) are sparse and high-dimensional, meaning related words are mathematically indistinguishable, resulting in a Cosine Similarity of 0.0 for all pairs.

**Semantic Blindness**

Because simple vectors fail to encode relationships, a model cannot "know" that a King is related to a Queen.

**100,000+**

**The Curse of Dimensionality**

One-Hot systems require vector sizes matching the entire vocabulary, leading to extreme memory requirements and computational inefficiency.

### The 3D Semantic Space

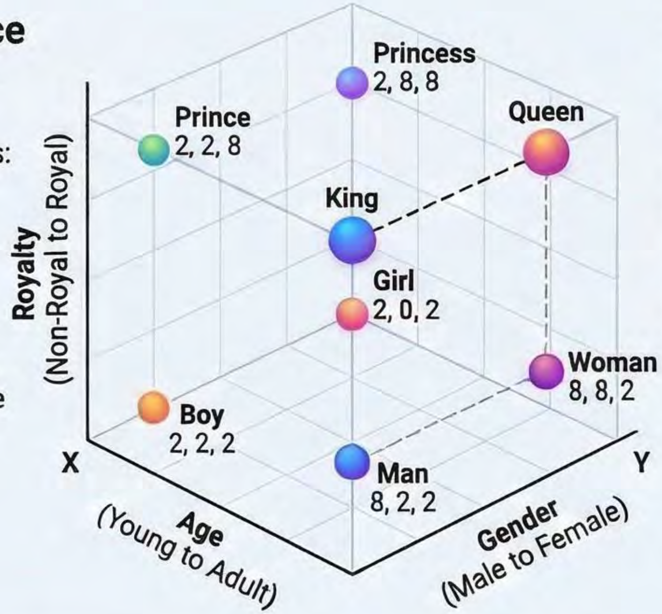
**Mapping Meaning to Axes**

Words are assigned coordinates across three primary semantic axes: Age, Gender, and Royalty.

**Decomposing a Word**

**King**

The word "King" is a point in space representing an Adult (Age: 8), Male (Gender: 2), and Royal (Royalty: 8) entity.




**Vector Map**

| Word | X | Y | Z |
|---|---|---|---|
| Boy | 2 | 2 | 2 |
| Girl | 2 | 8 | 2 |
| Man | 8 | 2 | 2 |
| Woman | 8 | 8 | 2 |
| Prince | 2 | 2 | 8 |
| Princess | 2 | 8 | 8 |
| King | 8 | 2 | 8 |

# King - Man + Woman = Queen

### Solving the Classic Analogy

**Step 1: King - Man**
(Isolating Royalty)

King (8, 2, 8)
- Man (8, 2, 2)

= (8-8, 2-2, 8-2)
= (0, 0, 6)

**Pure Royalty**

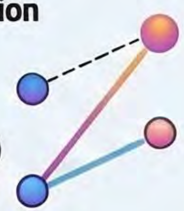
**Step 2: + Woman**
(Introducing Femininity)

(0, 0, 6) + Woman (8, 8, 2)

= (0+8, 0+8, 6+2)
= (8, 8, 8)

**The Final Equation**

King (8, 2, 8)
- Man (8, 2, 2)
+ Woman (8, 8, 2)
= **Queen (8, 8, 8)**



This spatial transformation lands perfectly on the pre-existing vector for Queen.

### Global Categorical Relationships

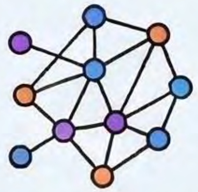
**Geography and Capitals**

Paris − France + Germany ≈ Berlin

**Verb Tense and Morphology**

Walk − Walked ≈ Swim − Swam

**Unsupervised Semantic Truths**



These relationships are derived purely from statistical co-occurrence, proving models can learn deep semantic truths without human-programmed rules.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# Directional Effects in Semantic Vector Arithmetic

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| Pattern | Calculations | Resulting Vector | Result | Inference |
|---|---|---|---|---|
| **King – Boy + Boy** | (2, 8, 8) – (2, 2, 2) + (2, 2, 2) | (2, 8, 8) | King | - |
| **King – Boy + Girl** | (2, 8, 8) – (2, 2, 2) + (8, 2, 2) | (8, 8, 8) | Queen | Royal Adult Female |
| **King – Boy + Man** | (2, 8, 8) – (2, 2, 2) + (2, 8, 2) | (2, 14, 8) | - | |
| **King – Boy + Woman** | (2, 8, 8) – (2, 2, 2) + (8, 8, 2) | (8, 14, 8) | - | |
| **King – Boy + Prince** | (2, 8, 8) – (2, 2, 2) + (2, 2, 8) | (2, 8, 14) | - | |
| **King – Boy + Princess** | (2, 8, 8) – (2, 2, 2) + (8, 2, 8) | (8, 8, 14) | - | |
| **King – Boy + King** | (2, 8, 8) – (2, 2, 2) + (2, 8, 8) | (2, 14, 14) | - | |
| **King – Boy + Queen** | (2, 8, 8) – (2, 2, 2) + (8, 8, 8) | (8, 14, 14) | - | |
| | | | | |
| **King – Girl + Boy** | (2, 8, 8) – (8, 2, 2) + (2, 2, 2) | (-4, 8, 8) | - | - |
| **King – Girl + Girl** | (2, 8, 8) – (8, 2, 2) + (8, 2, 2) | (2, 8, 8) | King | - |
| **King – Girl + Man** | (2, 8, 8) – (8, 2, 2) + (2, 8, 2) | (-4, 14, 8) | - | |
| **King – Girl + Woman** | (2, 8, 8) – (8, 2, 2) + (8, 8, 2) | (2, 14, 8) | - | |
| **King – Girl + Prince** | (2, 8, 8) – (8, 2, 2) + (2, 2, 8) | (-4, 8, 14) | - | |
| **King – Girl + Princess** | (2, 8, 8) – (8, 2, 2) + (8, 2, 8) | (2, 8, 14) | - | |
| **King – Girl + King** | (2, 8, 8) – (8, 2, 2) + (2, 8, 8) | (-4, 14, 14) | - | |
| **King – Girl + Queen** | (2, 8, 8) – (8, 2, 2) + (8, 8, 8) | (2, 14, 14) | - | |
| | | | | |
| **King – Man + Boy** | (2, 8, 8) – (2, 8, 2) + (2, 2, 2) | (2, 2, 8) | Prince | Royal Young Male |
| **King – Man + Girl** | (2, 8, 8) – (2, 8, 2) + (8, 2, 2) | (8, 2, 8) | Princess | Royal Young Female |
| **King – Man + Man** | (2, 8, 8) – (2, 8, 2) + (2, 8, 2) | (2, 8, 8) | King | - |
| **King – Man + Woman** | (2, 8, 8) – (2, 8, 2) + (8, 8, 2) | (8, 8, 8) | Queen | Royal Adult Female |
| **King – Man + Prince** | (2, 8, 8) – (2, 8, 2) + (2, 2, 8) | (2, 2, 14) | - | - |
| **King – Man + Princess** | (2, 8, 8) – (2, 8, 2) + (8, 2, 8) | (8, 2, 14) | - | - |
| **King – Man + King** | (2, 8, 8) – (2, 8, 2) + (2, 8, 8) | (2, 8, 14) | - | - |
| **King – Man + Queen** | (2, 8, 8) – (2, 8, 2) + (8, 8, 8) | (8, 8, 14) | - | - |
| | | | | |
| **King – Woman + Boy** | (2, 8, 8) – (8, 8, 2) + (2, 2, 2) | (-4, 2, 8) | - | - |
| **King – Woman + Girl** | (2, 8, 8) – (8, 8, 2) + (8, 2, 2) | (2, 2, 8) | Prince | Royal Young Male |
| **King – Woman + Man** | (2, 8, 8) – (8, 8, 2) + (2, 8, 2) | (-4, 8, 8) | - | |
| **King – Woman + Woman** | (2, 8, 8) – (8, 8, 2) + (8, 8, 2) | (2, 8, 8) | King | - |
| **King – Woman + Prince** | (2, 8, 8) – (8, 8, 2) + (2, 2, 8) | (-4, 2, 14) | - | |
| **King – Woman + Princess** | (2, 8, 8) – (8, 8, 2) + (8, 2, 8) | (2, 2, 14) | - | |
| **King – Woman + King** | (2, 8, 8) – (8, 8, 2) + (2, 8, 8) | (-4, 8, 14) | - | |
| **King – Woman + Queen** | (2, 8, 8) – (8, 8, 2) + (8, 8, 8) | (2, 8, 14) | - | |
| | | | | |
| **King – Prince + Boy** | (2, 8, 8) – (2, 2, 8) + (2, 2, 2) | (2, 8, 2) | Man | Non-Royal Adult Male |
| **King – Prince + Girl** | (2, 8, 8) – (2, 2, 8) + (8, 2, 2) | (8, 8, 2) | Woman | Non-Royal Adult Female |
| **King – Prince + Man** | (2, 8, 8) – (2, 2, 8) + (2, 8, 2) | (2, 14, 2) | - | - |
| **King – Prince + Woman** | (2, 8, 8) – (2, 2, 8) + (8, 8, 2) | (8, 14, 2) | - | - |
| **King – Prince + Prince** | (2, 8, 8) – (2, 2, 8) + (2, 2, 8) | (2, 8, 8) | King | - |
| **King – Prince + Princess** | (2, 8, 8) – (2, 2, 8) + (8, 2, 8) | (8, 8, 8) | Queen | Royal Adult Female |
| **King – Prince + King** | (2, 8, 8) – (2, 2, 8) + (2, 8, 8) | (2, 14, 8) | - | - |
| **King – Prince + Queen** | (2, 8, 8) – (2, 2, 8) + (8, 8, 8) | (8, 14, 8) | - | - |
| | | | | |
| **King – Princess + Boy** | (2, 8, 8) – (8, 2, 8) + (2, 2, 2) | (-4, 8, 2) | - | - |
| **King – Princess + Girl** | (2, 8, 8) – (8, 2, 8) + (8, 2, 2) | (2, 8, 2) | Man | Non-Royal Adult Male |
| **King – Princess + Man** | (2, 8, 8) – (8, 2, 8) + (2, 8, 2) | (-4, 14, 2) | - | |
| **King – Princess + Woman** | (2, 8, 8) – (8, 2, 8) + (8, 8, 2) | (2, 14, 2) | - | - |
| **King – Princess + Prince** | (2, 8, 8) – (8, 2, 8) + (2, 2, 8) | (-4, 8, 8) | - | |
| **King – Princess + Princess** | (2, 8, 8) – (8, 2, 8) + (8, 2, 8) | (2, 8, 8) | King | - |
| **King – Princess + King** | (2, 8, 8) – (8, 2, 8) + (2, 8, 8) | (-4, 14, 8) | - | |
| **King – Princess + Queen** | (2, 8, 8) – (8, 2, 8) + (8, 8, 8) | (2, 14, 8) | - | |
| | | | | |
| **King – King + Boy** | (2, 8, 8) – (2, 8, 8) + (2, 2, 2) | (2, 2, 2) | Boy | - |
| **King – King + Girl** | (2, 8, 8) – (2, 8, 8) + (8, 2, 2) | (8, 2, 2) | Girl | - |
| **King – King + Man** | (2, 8, 8) – (2, 8, 8) + (2, 8, 2) | (2, 8, 2) | Man | - |
| **King – King + Woman** | (2, 8, 8) – (2, 8, 8) + (8, 8, 2) | (8, 8, 2) | Woman | - |
| **King – King + Prince** | (2, 8, 8) – (2, 8, 8) + (2, 2, 8) | (2, 2, 8) | Prince | - |
| **King – King + Princess** | (2, 8, 8) – (2, 8, 8) + (8, 2, 8) | (8, 2, 8) | Princess | - |
| **King – King + King** | (2, 8, 8) – (2, 8, 8) + (2, 8, 8) | (2, 8, 8) | King | - |
| **King – King + Queen** | (2, 8, 8) – (2, 8, 8) + (8, 8, 8) | (8, 8, 8) | Queen | - |
| | | | | |
| **King – Queen + Boy** | (2, 8, 8) – (8, 8, 8) + (2, 2, 2) | (-4, 2, 2) | - | - |
| **King – Queen + Girl** | (2, 8, 8) – (8, 8, 8) + (8, 2, 2) | (2, 2, 2) | Boy | Non-Royal Young Male |
| **King – Queen + Man** | (2, 8, 8) – (8, 8, 8) + (2, 8, 2) | (-4, 8, 2) | - | - |
| **King – Queen + Woman** | (2, 8, 8) – (8, 8, 8) + (8, 8, 2) | (2, 8, 2) | Man | Non-Royal Adult Male |
| **King – Queen + Prince** | (2, 8, 8) – (8, 8, 8) + (2, 2, 8) | (-4, 2, 8) | - | |
| **King – Queen + Princess** | (2, 8, 8) – (8, 8, 8) + (8, 2, 8) | (2, 2, 8) | Prince | Royal Young Male |
| **King – Queen + King** | (2, 8, 8) – (8, 8, 8) + (2, 8, 8) | (-4, 8, 8) | - | - |
| **King – Queen + Queen** | (2, 8, 8) – (8, 8, 8) + (8, 8, 8) | (2, 8, 8) | King | - |

## The table demonstrates three important things very clearly:

- Some analogies land exactly on known semantic points
- Some produce valid numeric vectors, but outside the defined vocabulary
- The cleanest analogies happen when the operation modifies one semantic dimension in a structurally consistent way

- King – Boy + Girl → Queen ✓
- King – Girl + Boy → invalid ✗

This is a directional semantic system. This represents that removing a female vector from a male word produces a structurally unstable transformation unless corrected by another strong female component.

King - Woman = (-6, 0, 6) removes:

- **female component (strong negative shift)**
- **age completely (0 change)**
- keeps **royalty**

Subtracting **Woman** is not just removing "female" — it **over-corrects gender** and pushes the vector into a negative region. The embedding space is linear, but transformations are **direction-sensitive and not symmetric**, especially when subtracting strong components like "Woman."

King - Princess = (-6, 6, 0) removes:

- female component (**strong negative shift**)
- adds age (**young → adult**)
- keeps **non-royalty baseline (0 change in Z)**

**Changes multiple dimensions simultaneously.**

| Direction | Operation |
|---|---|
| **Gender** | King → Queen |
| **Age** | King → Prince |
| **Royalty** | King → Man |
| **Gender + Age** | King → Princess |
| **Gender + Royalty** | King → Woman |
| **Age + Royalty** | King → Boy |
| **Gender + Age + Royalty** | King → Girl |


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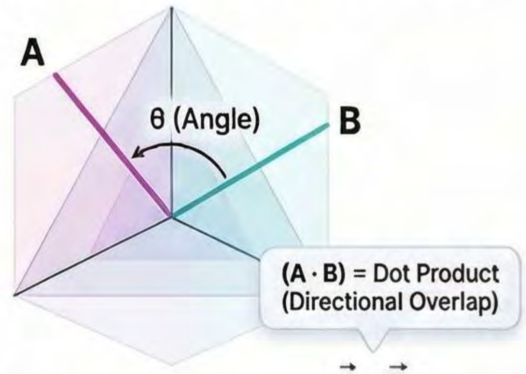
# **Cosine Similarity:** The Geometric Compass of Word Embeddings

A metric measuring semantic relationships by focusing on vector alignment (angle) rather than magnitude.

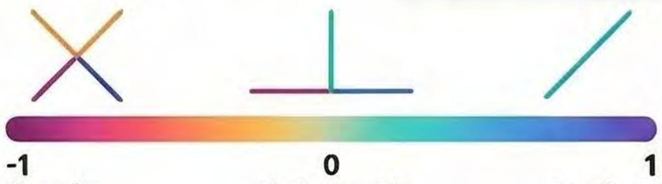
## THE MATHEMATICAL MECHANISM




$$\text{Cosine Similarity } (\cos\theta) = \frac{\vec{A}\cdot\vec{B}}{\|A\|\cdot\|B\|}$$

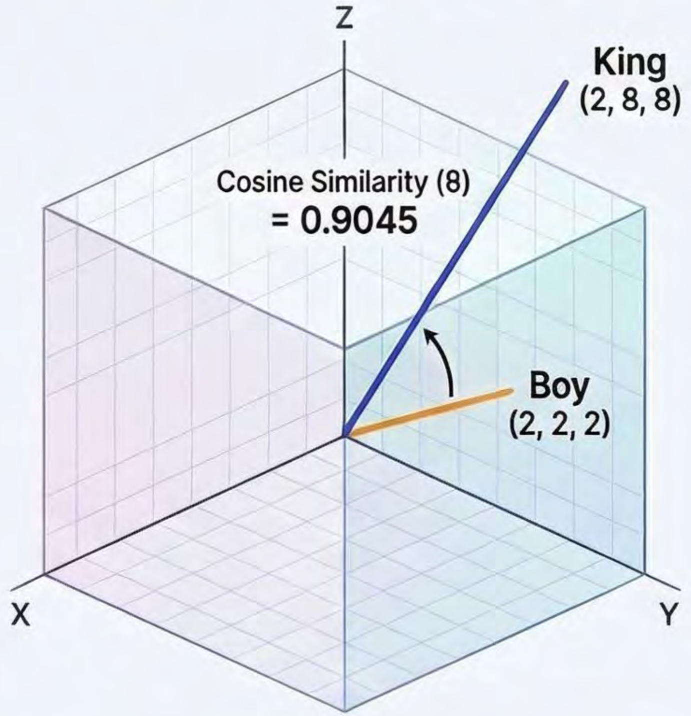
(||A|| · ||B||) =
Product of Magnitudes
(Length-Based Scaling)

### The [-1, 1] Interval




**Direction over Magnitude:** Measures the angle to isolate semantic alignment.

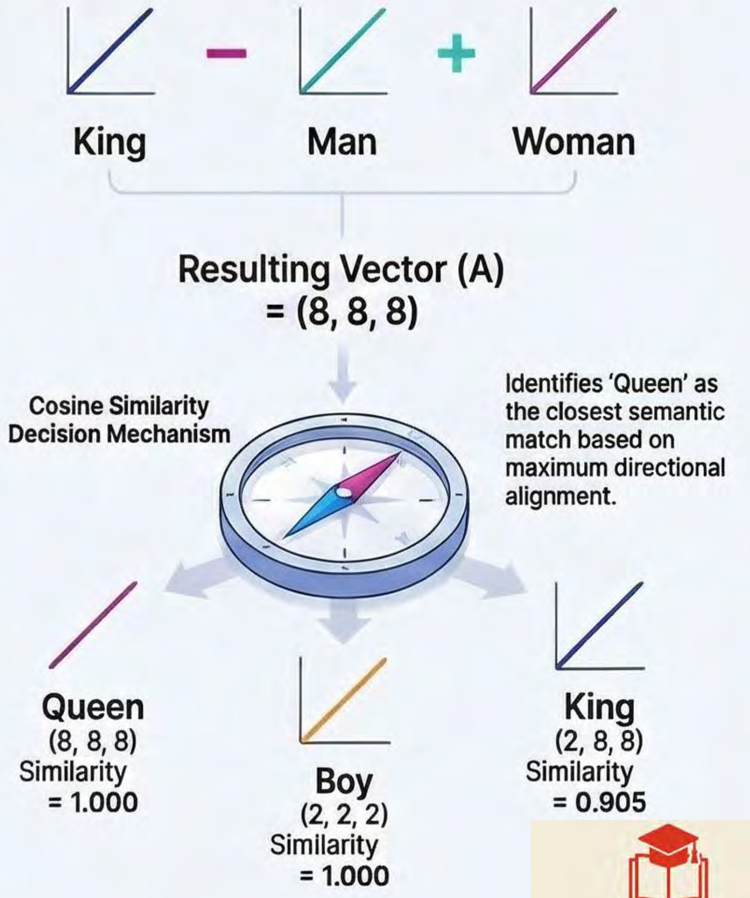
## COSINE SIMILARITY IN ACTION




**Comparing 'King' & 'Boy'**

High similarity due to shared structural alignment across dimensions, despite different magnitudes.

## THE 'QUEEN' ANALOGY: A GEOMETRIC DECISION MECHANISM





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# Cosine Similarity as the Decision Mechanism in Vector Analogies

| Word | Vector (B) | A.B [A - Result] | $\lVert A \rVert$ | $\lVert B \rVert$ | Cos(θ) |
|---|---|---|---|---|---|
| **Boy** | (2, 2, 2) | (2*8) + (2*8) + (2*8) = 48 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{2^2+2^2+2^2}\ \sqrt{12}$ | $\frac{48}{\sqrt{192}\sqrt{12}} = 1.000$ |
| **Girl** | (8, 2, 2) | (8*8) + (2*8) + (2*8) = 96 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{8^2+2^2+2^2} = \sqrt{72}$ | $\frac{96}{\sqrt{192}\sqrt{72}} = 0.816$ |
| **Man** | (2, 8, 2) | (2*8) + (8*8) + (2*8) = 96 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{2^2+8^2+2^2} = \sqrt{72}$ | $\frac{96}{\sqrt{192}\sqrt{72}} = 0.816$ |
| **Woman** | (8, 8, 2) | (8*8) + (8*8) + (2*8) = 144 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{8^2+8^2+2^2} = \sqrt{132}$ | $\frac{144}{\sqrt{192}\sqrt{132}} = 0.905$ |
| **Prince** | (2, 2, 8) | (2*8) + (2*8) + (8*8) = 96 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{2^2+2^2+8^2} = \sqrt{72}$ | $\frac{96}{\sqrt{192}\sqrt{72}} = 0.816$ |
| **Princess** | (8, 2, 8) | (8*8) + (2*8) + (8*8) = 144 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{8^2+2^2+8^2} = \sqrt{132}$ | $\frac{144}{\sqrt{192}\sqrt{132}} = 0.905$ |
| **King** | (2, 8, 8) | (2*8) + (8*8) + (8*8) = 144 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{2^2+8^2+8^2} = \sqrt{132}$ | $\frac{144}{\sqrt{192}\sqrt{132}} = 0.905$ |
| **Queen** | (8, 8, 8) | (8*8) + (8*8) + (8*8) = 192 | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\sqrt{8^2+8^2+8^2} = \sqrt{192}$ | $\frac{192}{\sqrt{192}\sqrt{192}} = 1.000$ |

When the resulting analogy vector A = (8,8,8) is compared directly with all word vectors, both “Boy” and “Queen” obtain a cosine similarity of 1.0 because both are perfectly collinear with the result. This occurs because cosine similarity measures only directional alignment and ignores magnitude. Since (8,8,8)=4.(2,2,2), “Boy” lies in exactly the same direction as the result vector.

If cosine similarity is computed on **directional vectors** rather than raw word vectors, the answer becomes much cleaner. For example, the desired gender transformation from “King” to its female counterpart is represented by the directional vector Queen-King=(6,0,0). Comparing this vector with candidate displacement vectors such as Woman-King, Princess-King, and Queen-King produces the strongest match for “Queen,” yielding a perfect cosine similarity of 1.0. Thus, directional cosine similarity provides a more semantically precise method for analogy resolution than direct comparison of raw word vectors.

Cosine similarity acts as the **decision mechanism** that translates vector arithmetic into interpretable linguistic outputs.


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p-Pathshala
A New Lesson Every Sunday
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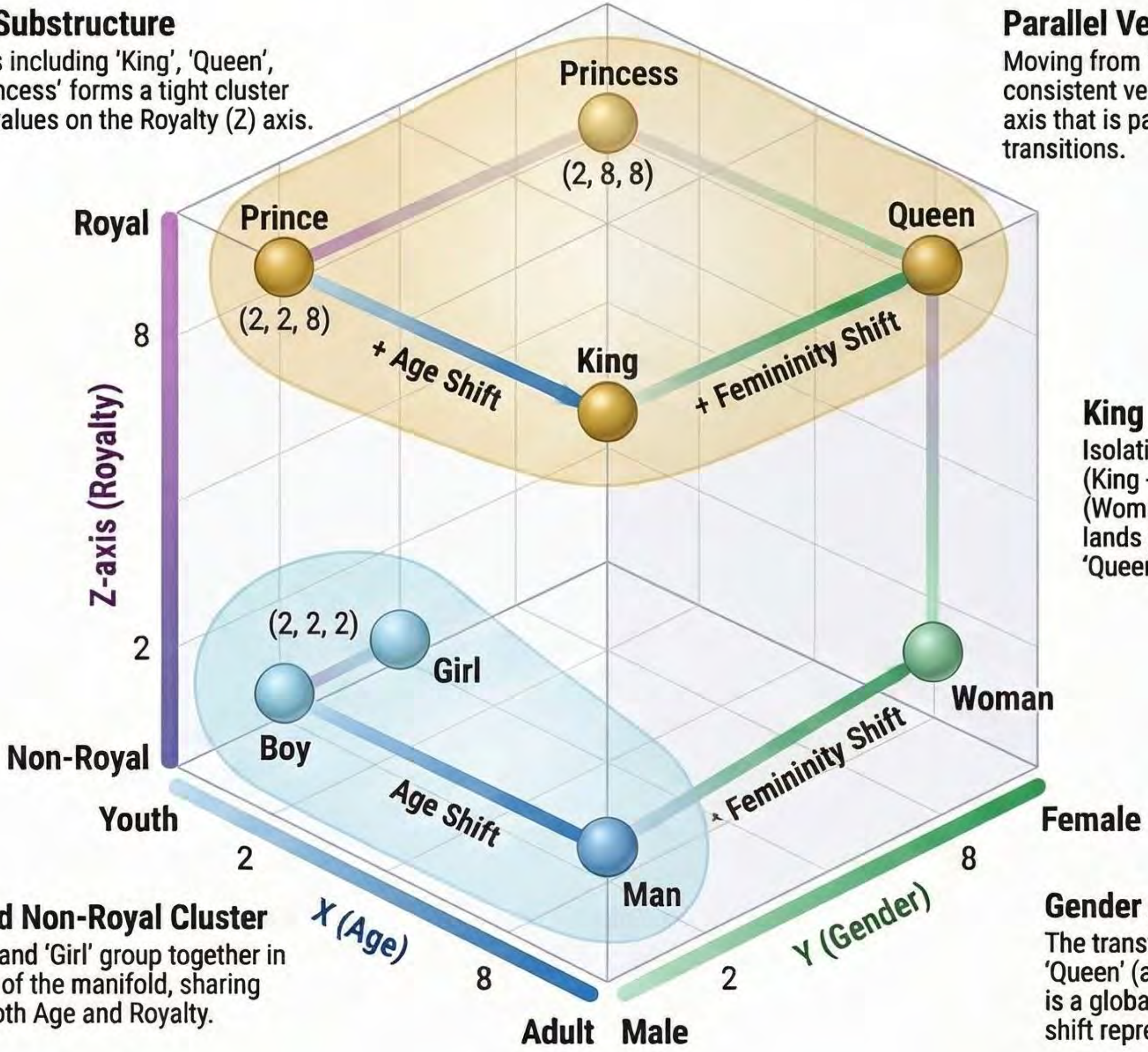
# The Manifold of Meaning: A Geometric Interpretation of Word Embeddings

### The Royalty Substructure
A group of words including 'King', 'Queen', 'Prince', and 'Princess' forms a tight cluster defined by high values on the Royalty (Z) axis.



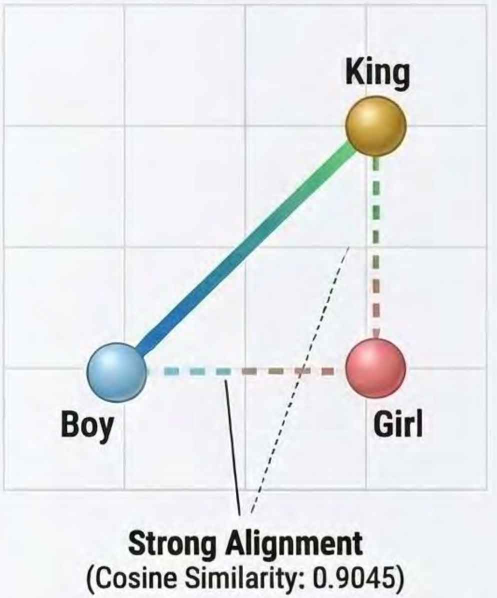

### The Youth and Non-Royal Cluster
Words like 'Boy' and 'Girl' group together in the lower region of the manifold, sharing low values for both Age and Royalty.

### Parallel Vector Transformations
Moving from 'Boy' to 'Man' follows a consistent vector direction along the Age axis that is parallel to other age-based transitions.

### King - Man + Woman = Queen
Isolating the 'Royalty' component (King - Man) and adding 'Femininity' (Woman) results in a vector that lands perfectly on the coordinate for 'Queen'.

### Gender Axis Consistency
The transformation from 'King' to 'Queen' (a shift from 2 to 8 on the Y-axis) is a globally consistent directional shift representing 'femininity'.

### The Latent Relationship Paradox




### Why 'Boy' is Closer to 'King' than 'Girl'
'Boy' (2, 2, 2) shares the Male directional component (Y=2) with 'King' (8, 2, 8). Cosine similarity captures this shared structural alignment across dimensions.

### Weak Alignment
'Girl' (2, 8, 2) differs on both the Gender and Royalty axes, placing it geometrically further from 'King'. Meaning is not a single label but distributed across multiple axes simultaneously.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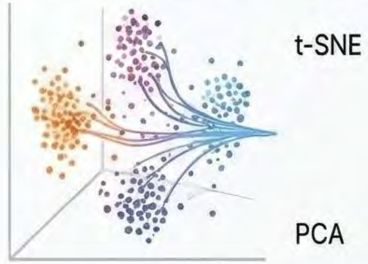
# Mapping the Mind of Machines: Semantic Clustering and Outlier Detection

Semantic spaces project words into geometric coordinates where related concepts form visual "islands". By calculating the distance between these points, algorithms can group similar items and identify "outliers" that do not belong to a specific group.

## The Geometry of Clustering




### Projecting Semantic Islands

Abstract dimensionality reduction algorithms like tSNE and PCA.



### Categorical Clustering

Related words (e.g., apple, orange) form fight "Iruit" clusters separate from "animal" clusters.

### Granular Proximity

Highly specific terms like "kitten" and "feline" exhibit microscopic distances within broader clusters.

## Outlier Detection and Validation


### The Centroid Calculation

Algorithms find the average "middle point" (centroid) for a list like ["cat", "dog", "mouse"].

### Flagging the Outlier

"Computer" is flagged because its cosine distance lies too far from the group centroid.

### Human Benchmark judgement

SimLex-999 WordSim-353

Model quality is validated using SimLex-999 or WordSim-353 to ensure correlation with human judgment.

### Data Table: Outlier Detection Logic

| Word List | Relationship to Centroid | Status |
|---|---|---|
| Cat, Dog, Mouse | Low Cosine Distance | In-Cluster |
| Computer | High Cosine Distance | Semantic Outlier |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